Flow Turbulence Combust
https://doi.org/10.1007/s10494-017-9867-1

VLES Modeling of Flow Over Walls with Variably-shaped Roughness by Reference to Complementary DNS

Benjamin Krumbein[1] · Pourya Forooghi[2] · Suad Jakirlić[1] · Franco Magagnato[2] · Bettina Frohnapfel[2]

Received: 29 March 2017 / Accepted: 12 October 2017
© Springer Science+Business Media B.V. 2017

Abstract Turbulent flow over variably-shaped rough walls, characterized by either a regular or a random arrangement of axisymmetric roughness elements in an open channel flow configuration, is investigated computationally within a VLES (Very Large Eddy Simulation) framework by utilizing a volumetric forcing-based roughness model. The prime objective of the present work is to assess the roughness model's capability to predict mean velocities and turbulent intensities in conjunction with this recently formulated hybrid LES/RANS (Reynolds-Averaged Navier-Stokes) model. The friction velocity-based Reynolds number is in the range $\mathrm{Re}_\tau = 460 - 500$. A non-dimensional drag function accounting for the shape of the roughness elements is introduced and evaluated based on the results of complementary direct numerical simulations (DNS). The dynamics of the residual motion of the presently adopted VLES methodology is described by an appropriately modified elliptic-relaxation-based $\zeta - f$ ($\zeta = \overline{v^2}/k$) RANS model.

Keywords DNS · Hybrid LES/RANS (VLES) · Elliptic relaxation eddy-viscosity model of turbulence · Volumetric-forcing-based roughness modeling · Open channel flow

✉ Benjamin Krumbein
krumbein@sla.tu-darmstadt.de

Suad Jakirlić
jakirlic@sla.tu-darstadt.de

1 Institute of Fluid Mechanics and Aerodynamics, Technische Universität Darmstadt, Alarich-Weiss-Straße 10, 64287 Darmstadt, Germany

2 Karlsruhe Institute of Technology, Institute of Fluid Mechanics, Kaiserstraße 10, 76131 Karlsruhe, Germany

Published online: 23 October 2017

1 Introduction

Turbulent flows over rough walls are frequently encountered in engineering and environmental applications. Examples include flows over aged turbine blades, accreting ice surfaces (at e.g. airplane wings and wind turbine blades) or plant and urban canopies in the atmospheric boundary layer. Furthermore, surface roughness, primarily due to depositions, plays an important role in internal combustion engines and exhaust systems, affecting pollutant output and efficiency.

To capture the effects of surface roughness in numerical simulations, the roughness geometry can be either directly considered, as for example by using immersed boundary methods or a body-fitted mesh, or an appropriate model is employed to mimic the effects of the roughness on the flow. The first approach, namely exact geometrical representation, is mainly used in the context of direct numerical simulation (DNS) or large eddy simulations (LES), see e.g. Bhaganagar et al. [1]. While this approach nominally provides accurate results, the increased computational demand of resolving the roughness topography typically limits the methodology to fairly simple geometries. For most practical applications the computational effort of resolving the roughness exceeds the available computational resources or the exact geometry of the surface topography is simply not known (e.g. roughness due to depositions in engines), necessitating the need for a roughness model. A commonly used approach to account for effects of surface roughness is appropriate extension of RANS (Reynolds-Averaged Navier-Stokes) models or wall boundary conditions based on the equivalent sand-grain roughness introduced by Nikuradse [2]. While practical, the reduction of the roughness effect onto a single parameter is not sufficient to provide reasonable results for the mean velocity and turbulent intensities in the vicinity of a rough wall, which for example might be necessary in the context of more complex computational fluid dynamics applications (as e.g. in near-wall reactive flows).

In the present work, the effect of the roughness is modeled by introducing a 'form drag' term in the Navier-Stokes equations. This approach, referred to as discrete element approach by Taylor et al. [3], proposed originally for two-dimensional turbulent boundary layers, was later extended to three-dimensional flows, see e.g. Miyake et al. [4] or Busse et al. [5]. For comparison purposes and for determination of the relevant model functions a complementary DNS database is used, where the roughness geometry is resolved using an immersed boundary method (IBM), Forooghi et al. [6]. Figure 1 gives an overview of the five roughness topographies considered in this work. Open channel flow over a rough wall at $\mathrm{Re}_\tau \approx 460 - 470$ is initially investigated for different roughness geometries consisting

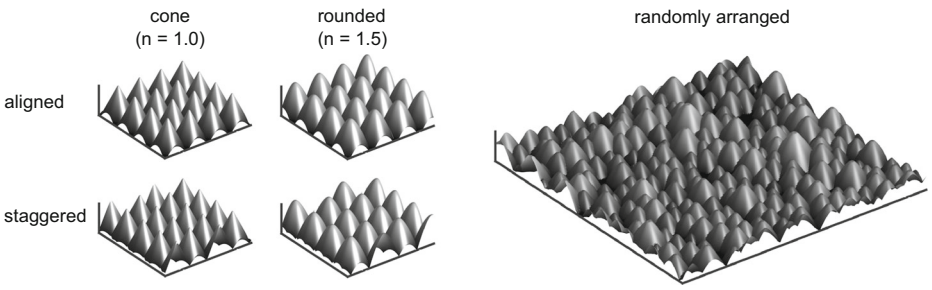


Fig. 1 Considered roughness geometries: the four regularly-arranged roughness topographies on the left, randomly-arranged roughness on the right

of a regular arrangement of axisymmetric roughness elements (Fig. 1-left). The roughness height is $h^+ \approx 49$. Two arrangements (aligned/staggered) and two element shapes ($n = 1.0$, $n = 1.5$) are considered. Subsequently, the roughness modeling approach is applied to a more realistic rough surface, which is generated by randomly distributing axisymmetric roughness elements of varying shape and height on a smooth reference surface (Fig. 1-right). Admittedly, the roughness elements used in this work cannot be ultimately considered as a geometrically true representation of the features found in naturally formed roughness morphologies. On the other hand, the use of these types of symmetric roughness elements as 'effective' reproduction of realistic roughness is a common practice in the literature (see for example Tarada [7] and Stripf et al. [8]). Therefore, as long as the 'proof-of-concept' is intended, it is believed that the use of roughness elements can lead to reasonable results as long as the main topographical properties of the roughness are practical. The skewness and kurtosis of the surface height distribution (two widely used morphological parameters to describe rough surfaces in the literature) for the randomly-arranged roughness sample used in the work are $Sk = 0.21$ and $Ku = 2.62$ respectively, which both agree well with the values measured for commonly used gas turbine blades (see e.g. Bons et al. [9]). Also, the effective slope (mean streamwise surface slope) for this surface is equal to $ES \approx 0.88$, which is high enough to ensure that form drag due to roughness elements is the dominant force; therefore the quadratic relation between the drag force and velocity holds.

The objective of this work is to assess the roughness model's capability of predicting the mean velocity profiles and turbulent intensities in conjunction with a newly-formulated hybrid RANS/LES model, based nominally on the VLES concept of Speziale et al. [10]. Unlike in the LES framework utilizing most commonly the Subgrid-Scale (SGS) models of the Smagorinsky type (a 0-equation model), where the grid spacing Δ represents the characteristic size of the largest unresolved scales (subgrid-scales), the representative length (and time) scales of the residual turbulence entering the time-dependent equations of motion in the present VLES and some relevant hybrid LES/RANS methodologies, such as PITM (Partially-Integrated Transport Model, Schiestel and Dejoan [11]) and PANS (Partially-Averaged Navier-Stokes, Basara et al. [12]), are determined by solving respective equations describing the dynamics of corresponding kinetic energy of turbulence and its dissipation rate. Consequently, the grid spacing Δ incorporated into the Hybrid LES/RANS models, represents just one of several model parameters serving for the determination of the turbulent viscosity. The relevant model formulations point to complex relationships involving different turbulent quantities exhibiting high level of coherence. Such a model rationale has inherently much more physics, allowing in general the use of coarser spatial and temporal resolutions, representing the feature being especially beneficial at high flow Reynolds numbers. Common feature of these, as well as of all scale-resolving modeling strategies is appropriate suppression of the fully-modeled RANS turbulence towards a corresponding sub-scale level (analog to the sub-grid scales in the LES framework). Whereas an appropriate dissipation level enhancement and consequent turbulence intensity reduction in both PANS and PITM methods is achieved by reducing selectively the destruction term in the model dissipation equation by making the corresponding model coefficient grid-spacing-dependent, the present VLES is based on suppressing directly the turbulent viscosity to a corresponding sub-scale level in the equation governing the velocity field. This is enabled by introducing a grid-spacing dependent function controlling the fraction of the unresolved turbulence. Certain advantage of the VLES method is that there is no need for time/ensemble-averaged parameters in course of the computation, but only for directly available instantaneous turbulent flow quantities.

2 Computational Methodology

The incompressible flow field is obtained by solving the continuity and Navier-Stokes equations. A generalized (correspondingly filtered) form of these equations, applicable to both DNS and VLES, is given by

$$\frac{\partial \widetilde{U}_i}{\partial x_i} = 0 \tag{1}$$

and

$$\frac{\partial \widetilde{U}_i}{\partial t} + \widetilde{U}_j \frac{\partial \widetilde{U}_i}{\partial x_j} = -\frac{1}{\rho}\frac{\partial \widetilde{p}}{\partial x_i} + \frac{\partial}{\partial x_j}\left(\nu \frac{\partial \widetilde{U}_i}{\partial x_j} - \tau_{ij}\right) + \widetilde{f}_i \tag{2}$$

with $\widetilde{U}_i$ representing either the fully-resolved (DNS) or the appropriately filtered (VLES) velocity field. p is the pressure, ρ is the density and ν the coefficient of kinematic viscosity of the fluid. As usual in the LES-relevant framework, a spatial filtering operation denoted by the tilde operator $\widetilde{(\cdot)}$ is applied. The filtering operation gives rise to the additional term τ_{ij}, proportional nominally to the grid spacing, representing the influence of the residual (sub-filter) motion on the filtered velocity field. For a correspondingly large grid size, that is a sufficiently small filter width, the DNS limit is recovered: the filtered quantities become the fully-resolved quantities associated with vanishing stress tensor - $\tau_{ij} = 0$. However, in the VLES framework, described in more detail in Section 2.1, the residual stress tensor does not vanish and has to be modeled. The volumetric force term $\widetilde{f}_i$ in Eq. 2 serves different purposes in the VLES and DNS methodology: in the VLES framework the volumetric force term is introduced to model the blockage and form drag the roughness elements exert on the flow (see Section 2.2). In the DNS $\widetilde{f}_i$ represents the source term of the IBM method, enforcing no-slip boundary conditions at the surface of the roughness elements. Since blockage effects are included in the forcing term in both the volumetric forcing approach and the IBM method, no modifications to the continuity (1) is made.

2.1 Very large Eddy simulation

In the context of an eddy-viscosity-based modeling concept the residual stress tensor τ_{ij} in Eq. 2, associated with the non-resolved eddy motion, is linearly related to the filtered rate of the strain tensor $\widetilde{S}_{ij}$, leading to

$$\tau_{ij} = -2\nu_t \widetilde{S}_{ij} + \frac{1}{3}\tau_{kk}\delta_{ij}, \tag{3}$$

where $\nu_t = F_r \nu_t^{RANS}$, formulated in line with the proposal by Speziale [10], is the turbulent viscosity of the unresolved motion and ν_t^{RANS} is the one predicted by the background RANS-based sub-scale model, relying on the underlying unsteady flow field. F_r is the so-called 'resolution function', damping the influence of the RANS model to an appropriate level, depending on the local mesh resolution. The VLES-related sub-scale viscosity is consequently defined as

$$\nu_t = F_r C_\mu^\zeta \zeta_{us} k_{us} T_{us}. \tag{4}$$

The $()_{us}$-quantities are determined by employing the elliptic-relaxation-based RANS model of Hanjalić et al. [13], denoted by ζ-f. In addition to the transport equations for the turbulent kinetic energy k_{us} and its dissipation rate ε_{us}, the equations for the variable ζ_{us} $(= \overline{v^2}_{us}/k_{us})$,

representing a measure for the wall-normal turbulent intensity and the elliptic function f_{us} are solved. The model transport equations read

$$\frac{\partial k_{\mathrm{us}}}{\partial t} + \widetilde{U}_j \frac{\partial k_{\mathrm{us}}}{\partial x_j} = P_k - \varepsilon_{\mathrm{us}} + \frac{\partial}{\partial x_j}\left[\left(\nu + \frac{\nu_{\mathrm{t}}}{\sigma_k}\right)\frac{\partial k_{\mathrm{us}}}{\partial x_j}\right], \tag{5}$$

$$\frac{\partial \varepsilon_{\mathrm{us}}}{\partial t} + \widetilde{U}_j \frac{\partial \varepsilon_{\mathrm{us}}}{\partial x_j} = \frac{C_{\varepsilon_1} P_{\mathrm{us}} - C_{\varepsilon_2}\varepsilon_{\mathrm{us}}}{T_{\mathrm{us}}} + \frac{\partial}{\partial x_j}\left[\left(\nu + \frac{\nu_{\mathrm{t}}}{\sigma_\varepsilon}\right)\frac{\partial \varepsilon_{\mathrm{us}}}{\partial x_j}\right], \tag{6}$$

$$\frac{\partial \zeta_{\mathrm{us}}}{\partial t} + \widetilde{U}_j \frac{\partial \zeta_{\mathrm{us}}}{\partial x_j} = f_{\mathrm{us}} - \frac{P_{\mathrm{us}}}{k_{\mathrm{us}}}\zeta_{\mathrm{us}} + \frac{\partial}{\partial x_j}\left[\left(\nu + \frac{\nu_{\mathrm{t}}}{\sigma_\zeta}\right)\frac{\partial \zeta_{\mathrm{us}}}{\partial x_j}\right], \tag{7}$$

$$L_{\mathrm{us}}^2 \nabla^2 f_{\mathrm{us}} - f_{\mathrm{us}} = \frac{1}{T_{\mathrm{us}}}\left(C_1 + C_2 \frac{P_{\mathrm{us}}}{\varepsilon_{\mathrm{us}}}\right)\left(\zeta_{\mathrm{us}} - \frac{2}{3}\right), \tag{8}$$

with the production rate $P_k = \nu_{\mathrm{t}} \widetilde{S}_{ij}\widetilde{S}_{ij}$, the time scale

$$T_{\mathrm{us}} = \frac{k_{\mathrm{us}}}{\varepsilon_{\mathrm{us}}} \tag{9}$$

and length scale

$$L_{\mathrm{us}} = C_L \max\left[\frac{k_{\mathrm{us}}^{3/2}}{\varepsilon_{\mathrm{us}}}, C_\eta \left(\frac{\nu^3}{\varepsilon_{\mathrm{us}}}\right)^{1/4}\right]. \tag{10}$$

The model coefficients are summarized in Table 1.

For the contribution function the formulation of Chang et al. [14], providing a seamless blending between the limits of DNS ($F_{\mathrm{r}} \to 0$) and RANS ($F_{\mathrm{r}} \to 1$) is adopted. The formulation originates from the proportionality between the length scales characterizing the underlying grid resolution and unresolved turbulence, resulting in a corresponding ratio as follows:

$$F_{\mathrm{r}} = \min\left[\left(\frac{\Delta}{\Lambda_{\mathrm{us}}}\right)^{4/3}, 1\right]. \tag{11}$$

Here, $\Delta = (\Delta_x \Delta_y \Delta_z)^{1/3}$ is the representative grid spacing and $\Lambda_{\mathrm{us}} = k_{\mathrm{us}}^{3/2}/\varepsilon_{\mathrm{us}}$ is the turbulent length scale.

Presently, no modifications to the underlying RANS model are made to account for the influence of the surface roughness on the sub-scale dynamics. This restricts the present model to meshes where a sufficiently large fraction of the turbulent kinetic energy can be resolved and hence the influence of the roughness on the sub-scale dynamics is comparatively small. To retain the VLES model's applicability in the RANS limit, appropriate source terms in the model equations have to be introduced.

The described model is implemented in the open-source continuum mechanics library OpenFOAM®. Structured hexahedral meshes are used. In line with standard practices for LES in the context of the finite volume method, second-order schemes are applied to prevent high numerical diffusion. The central differencing scheme is used for the spatial discretization of both the convective and diffusive term in the momentum equation. The second-order backward differentiation formula is adopted for the temporal discretization.

Table 1 Model coefficients in the present VLES model

C_μ^ζ	C_{ε_1}	C_{ε_2}	C_1	C_2	σ_k	σ_ε	σ_ζ	C_L	C_η
0.22	$1.4(1 + 0.045/\sqrt{\zeta})$	1.9	0.4	0.65	1	1.3	1.2	0.36	85

The present VLES model has been intensively validated in the past in a variety of different generic and complex flows subjected to separation, swirl and mean compression (Chang, 2014, 2016, [14, 15]), including also the flow over car configurations of different complexity, (Jakirlic et al., 2016, [16]).

2.2 Roughness treatment in the VLES framework

To account for the effect of surface roughness, a volumetric forcing approach similar to Busse et al. [5] is adopted. The model is based on the idea that the influence of surface roughness on the flow can be described with an additional drag term introduced in the momentum equation, accounting for the form drag and blockage effects the roughness elements exert on the

$$\rho \widetilde{f_i} = -\frac{1}{2}\,\rho\,\widetilde{U}_j\widetilde{U}_j \frac{\widetilde{U}_i}{\sqrt{\widetilde{U}_k\widetilde{U}_k}}\,\frac{D(y)}{L}. \tag{12}$$

The similarity to a drag force definition is obvious, with the dynamic pressure $\frac{1}{2}\,\rho\,\widetilde{U}_j\widetilde{U}_j$ and the orientation of the force acting against the flow direction. Instead of a drag coefficient, the non-dimensional drag function $D(y)$ accounting for the shape of the roughness elements is introduced. For dimensional reasons the introduction of a relevant length-scale L is necessary, which represents usually a volume to surface ratio of the roughness. Here the maximum roughness height h is adopted ($L = h$).

The key issue is the modeling of the drag function to length scale ratio. Different approaches involving volumetric forcing are proposed in the literature. If the mesh is sufficiently fine, the volumetric force can be chosen in a way forcing the velocity inside the roughness elements to take zero values. This approach however has similarly high requirements in terms of the mesh size as immersed boundary methods and, in addition, the shape of the roughness has to be prescribed. A more practical approach is to use a drag function $D(y)$ depending on the wall distance y, resulting in a constant drag coefficient in layers parallel to the wall. Commonly, constant values for D, independent of the wall distance, are prescribed in the roughness-affected layer (see e.g [3, 4, 17–19]). These values are usually either based on experiments specific to a given roughness topography or the values are chosen in a way to fit experimental data. Busse et al. [5] did not apply constant values but rather investigated the influence of several arbitrarily selected shape functions fulfilling several realizability constraints in a parametric study. They subsequently applied one of the adopted shape functions to the flow in a rod-roughened channel, investigated by means od DNS by Ashravian et al. (2004, [20]), with the roughness arrangement consisting of two-dimensional transverse square bars (resembling a 'k-type' roughness). The magnitude of the forcing function was appropriately tuned to reproduce the velocity profile in the log-law region. In the present work, the drag function profiles are determined based directly on the complementary DNS database.

Consistent with the model assumption of constant drag coefficients in layers parallel to the wall, superficial spatial averaging of the DNS data sets in steamwise and spanwise directions is performed. Subsequently, the streamwise component of the RANS equation governing the time-averaged quantities in the presently considered channel flow is used to calculate the volumetric force $\overline{f}_x$. This yields

$$\overline{f}_x = \frac{1}{\rho}\frac{\partial \overline{p}}{\partial x} - \frac{\partial}{\partial y}\left(\nu\frac{\partial \overline{U}}{\partial y} - \overline{uv}\right), \tag{13}$$

where $\partial\overline{p}/\partial x$ is the pressure gradient, $\overline{U}$ the velocity component in streamwise direction and $\overline{uv}$ the Reynolds shear-stress component, all taken as input variables from the respective DNS data set. The drag function is subsequently obtained by dividing the average volumetric force and length scale by the dynamic pressure, based on the mean velocity profile:

$$D(y) \approx -\frac{\overline{f}_x h}{\frac{1}{2}\rho\overline{U}^2}. \tag{14}$$

This approach is certainly not universally applicable for practical applications either, since DNS computations of the roughness topography have to be performed in advance to determine the model functions. On the other hand, the presently employed roughness modeling concept represents actually a kind of a 'proof-of-concept' procedure. The addition of a volumetric forcing term into the equation of motion, resembling the form of a drag force, represents certainly a very practical modeling approach. The drag function $D(y)$ is determined in the 'data-driven' manner using the results of the complementary DNS simulations. The respective database was finally used as an input into the VLES computations applied in conjunction with the 'drag-force-like' roughness model. Accordingly, the goal here is rather to investigate the limitations of the proposed model and to subsequently use the obtained data as a basis for an empirical model, which relates more closely the drag function to the geometrical properties of the roughness.

2.3 Direct numerical simulation

The pseudo-spectral incompressible Navier-Stokes solver SIMSON (see Chevalier et al. [21]) is used to carry out the DNS simulations. For the discretization in wall-parallel directions Fourier series are used, while Chebyshev series are employed in wall-normal direction. Time marching of the advective and forcing terms is realized by applying the explicit third order Runge-Kutta scheme. For the viscous terms the second order accurate implicit Crank-Nicolson scheme is adopted. A CFL number of 0.8 is applied in the current simulations. To obtain time-averaged statistics, averaging is carried out for at least 20 flow-through-times after statistical steady state is reached.

An immersed boundary method (IBM) based on Goldstein et al. [22] was implemented to numerically represent the exact geometry of the roughness elements. The method is based on imposing the no slip-condition at the fluid-solid interface by adding two source terms to the momentum equation containing instantaneous local velocity and its time integral. Due to use of IBM, fairly fine grids are necessary in the simulations. In order to ensure the grid-independence of those statistics discussed in this paper, a grid study is carried out. Figure 2 presents the profiles of mean streamwise velocity and all components of rms fluctuating velocities for one of the studied geometries in this study - the random roughness sample introduced in details in Section 2. Four grids are used in this grid study: Grid1 (N_x =768, N_y = 301, N_z =384), Grid2 (1152, 301, 576), Grid3x (1728, 301, 864) and Grid3y (1152,451, 576). Results indicate that first and second order one-point statistics are satisfactorily grid-independent even on the coarsest grid. This DNS/IBM methodology was validated against DNS data by Chan-Braun et al. [23], who applied a high fidelity solver on an extremely fine grid (Δx^+, Δy^+, $\Delta z^+ < 1$) to the flow over a surface with spherical roughness elements. It was found that both first and second order statistics agree well with the reference data base.

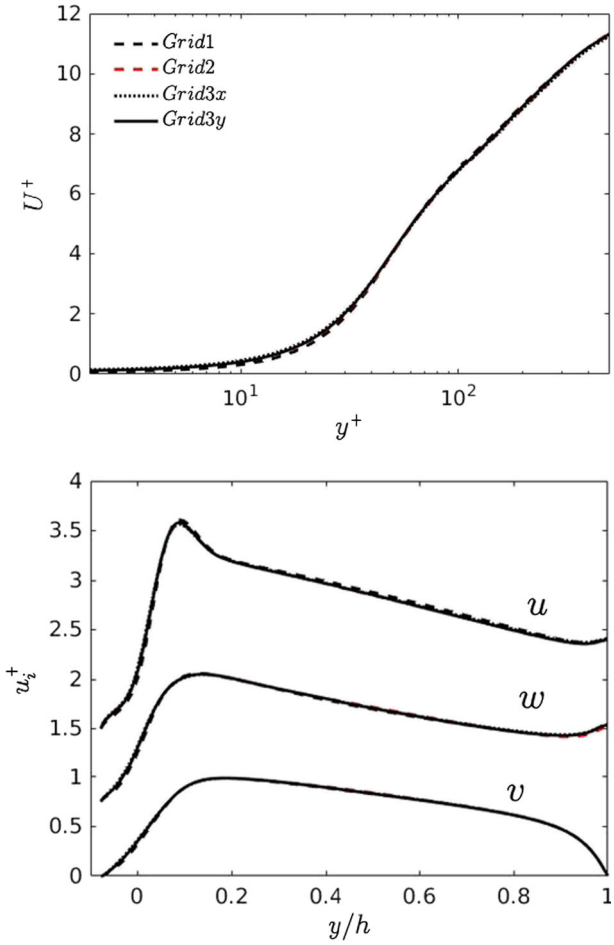


Fig. 2 Profiles of the mean velocity and turbulence intensity components obtained by the present DNS for different grid resolutions. The results relate to the randomly-arranged roughness morphology

3 Roughness Topography and Flow Properties

An open channel of height H with a rough bottom wall is employed as the test case in this work. Two types of roughness element distributions are considered: first, a regular arrangement of axisymmetric roughness elements is considered and secondly, similarly-shaped roughness elements of varying shape and size are distributed randomly, in order to generate a more realistic rough surface with prescribed statistical parameters.

The maximum height of the regularly arranged roughness layer is $h = 0.1H$. The shape of the axisymmetric roughness elements is given by

$$y = h \min\left[1 - \left(\frac{r_0 - r}{r_0 - R_0}\right)^n, 1\right], \tag{15}$$

where r is the radial coordinate of the local coordinate system with the origin defined by the intersection of the symmetry axis with the plane $y = 0$. Furthermore, R_0 and $r_0 = 0.05\, R_0$

are the radii at $y = 0$ and $y = h$ respectively and n is a parameter defining the shape of the surface. The roughness elements in the regularly arranged cases (see Fig. 1, left) are obtained with $n = 1.0$ (cone shaped) and $n = 1.5$ (rounded shape). Furthermore, aligned and staggered arrangements are considered. The distribution of the roughness elements, given by their distance δ_0 as well as their diameter is chosen in a way, where the solidity λ_f, defined as the ratio of the total projected frontal area of the roughness elements to the wall-normal projected area, is similar ($\lambda_f \approx 0.64$) in all cases. The geometrical parameters for the four regularly arranged roughness topographies are summarized in Table 2.

The randomly arranged rough surface is generated by placing roughness elements, the shape of which is described by Eq. 15, with variable height h and shape parameter n in a way, providing that certain prescribed statistical quantities are achieved. For a more detailed description of the generation approach for randomly arranged rough surface we refer to Forooghi et al. [6]. The random roughness considered in this work has Skewness Sk = 0.21, Kurtosis Ku = 2.6, solidity $\lambda_f = 0.45$, effective Slope ES = 0.88 and height $h_{rms}/H = 0.0417$.

The height of the channel available to the flow is reduced by the presence of the roughness elements. Therefore, an effective height H_{eff}, depending on the roughness topography has to be adopted as the length scale for the determination of the Reynolds number. Commonly, this effective height is $H_{eff} = H - y_0$, where y_0 is the position of the 'virtual wall', which can not be that clearly defined for the rough-wall case. For the determination of y_0 generally a posteriori methods are employed which often involve fitting the mean velocity profile to the logarithmic law of the wall, see e.g. Chan-Braun et al. [23] for an overview of definitions used in the literature. In this work a more practical a priori definition is preferred: the melt-down thickness y_{md}, i.e. the ratio of the volume of the roughness elements to the wall-normal projected area, is adopted to define the position of the virtual wall. This leads to the following expression for the effective channel height:

$$H_{eff} = H\left(1 - \frac{y_{md}}{H}\right). \tag{16}$$

The bulk Reynolds number is defined as $Re_b = U_b H_{eff}/\nu$, where U_b is the mean velocity based on the effective height H_{eff}. For the friction Reynolds number defined as $Re_\tau = U_\tau H_{eff}/\nu$, the friction velocity U_τ has to be determined. Since the immediate wall vicinity is not resolved, U_τ is estimated based on the extrapolation of the total shear-stress from a region away from the wall, where it varies linearly, to the position of the virtual wall, yielding

$$U_\tau = \left(\frac{\tau_{tot}}{1 - \frac{y - y_{md}}{H_{eff}}}\right)^{1/2}. \tag{17}$$

This approach is in fact equivalent to the estimation of the friction velocity based on a global momentum balance. For a full discussion on different approaches to the determination of the wall shear stress we refer to Chan-Braun et al. [23].

Table 2 Base radius R_0 and distance of the roughness elements δ_0, both non-dimensionalized by the roughness element height h for the four regular roughness configurations

	R_0/h	δ_0/h
Alig., $n = 1.0$	0.500	0.800
Alig., $n = 1.5$	0.555	0.888
Stag., $n = 1.0$	0.525	0.800
Stag., $n = 1.5$	0.583	0.888

Table 3 Melt-down thickness y_{md} non-dimensionalized with the channel height H as well as bulk and friction Reynolds numbers from DNS of the five considered rough cases and the smooth reference case

	y_{md}/H	$\mathrm{Re_b}$	$\mathrm{Re_\tau}$
Smooth	0	9064	500.7
Alig., $n = 1.0$	0.041	4822	470.9
Alig., $n = 1.5$	0.052	5236	463.5
Stag., $n = 1.0$	0.045	4498	467.6
Stag., $n = 1.5$	0.056	4819	460.9
Random	0.072	4051	498.1

Relative melt-down thickness as well as bulk and friction Reynolds numbers for all five considered roughness topographies are summarized in Table 3.

4 Computational Set-up

A sketch of the computational domain is shown in Fig. 3. In the DNS simulations, the dimensions are $L_x \times L_y \times L_z = 8H \times H \times 4H$ in streamwise, wall-normal and spanwise directions respectively, while in the VLES simulations a channel size of $4H \times H \times 2H$ was found to be sufficient, since no significant differences in the profiles of the mean velocity and turbulent intensities could be observed compared to the larger domain. No-slip boundary conditions are used at the plane $y = 0$ and, in the case of DNS/IBM simulations at the surface of the roughness elements. At the top boundary shear-free, non-permeable conditions are imposed. In streamwise and spanwise directions periodic boundary conditions are applied. In the DNS the flow is driven by a constant pressure gradient corresponding to the respective friction-velocity-based Reynolds number (see Table 3). In the VLES cases, the driving pressure gradient is adapted in each time step to ensure the correct bulk Reynolds number obtained by DNS. In all cases adaptive time stepping was used, keeping the maximum Courant-Friedrichs-Lewy (CFL) number below 0.8.

In order to determine the mesh resolution necessary for the DNS to achieve grid independent first and second order one-point statistics in combination with the implemented IBM method, a mesh dependence study was conducted. Based on this study a grid with $768 \times 301 \times 384$ cells is employed for the regular roughness cases and $1152 \times 301 \times 576$ cells for the random roughness, resulting in approximately 89 million and 200 million cells, respectively. This corresponds to a non-dimensional grid spacing of $\Delta x^+ = \Delta z^+ \approx 5$ and $\Delta y^+_{\mathrm{max}} \approx 3.5$ for the regular roughness cases and $\Delta x^+ = \Delta y^+_{\mathrm{max}} = \Delta z^+ \approx 3.5$ for the random roughness case. With this mesh resolution, each roughness element is discretized with at least $15 \times 60 \times 15$ grid points, ensuring a correct geometrical representation of the elements shape by the IBM method. The mesh size employed for the VLES computations

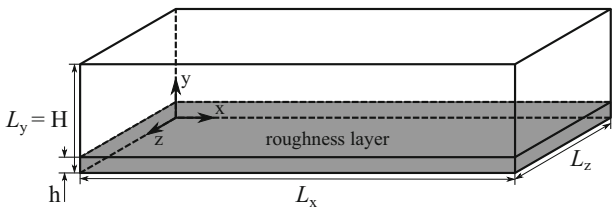


Fig. 3 Sketch of the computational domain

of all considered roughness cases is considerably smaller with $80 \times 64 \times 100$ cells totaling a number of 512 thousand cells. The mesh is homogeneous in streamwise and spanwise directions, while in wall-normal direction a grading towards the base wall is introduced. The grid spacing corresponds to $\Delta x^+ = 25$, $\Delta y^+ = 2 - 20$, $\Delta z^+ = 12.5$. A coarser grid resolution (preserving the number of cells in the wall-normal direction and halving them in both wall-parallel directions) has also been checked (the results are not shown here). The results show no noticeable difference with respect to the mean flow. Only important difference is in the somewhat overpredicted peak value of the streamwise component of the turbulence intensity within the roughness layer and, consequently, in the correspondingly reduced normal-to-the-wall and spanwise turbulence intensity components (due to an adequate 'energy redistribution' among them). Appropriate increase in the grid size, towards the currently adopted one, contributed to adequate capturing of the somewhat finer eddy structures originating from the roughness elements and, finally, to an appropriately reduced separation between the Reynolds stress components, in good agreement with the DNS reference. The corresponding ratio of the modelled turbulence to the total one - k_{sgs}/k_{total}, being actually proportional to the resolution function F_r, reduces gradually from about 95% within the roughness layer to about 5% in the region above the roughness crest, increasing slightly to less than 10% in the remaining part of the cross-section, Fig. 4.

5 Results and Discussion

5.1 Regularly arranged roughness

To obtain the drag functions for the four considered regular roughness topographies, Eqs. 13 and 14 are evaluated based on the input originating from the respective DNS data sets. The resulting profiles, exhibiting high degree of scalability, are displayed in Fig. 5. For numerical stability reasons the values of $D(y)$ are bound to a maximum value of 200 in the lower part of the roughness layer. These large values of D, which have to be bounded, are produced by Eq. 14 due to the mean velocity, whose square appears in the denominator, approaching zero values in this region (partly due to blockage effects). Since the mean velocity is small,

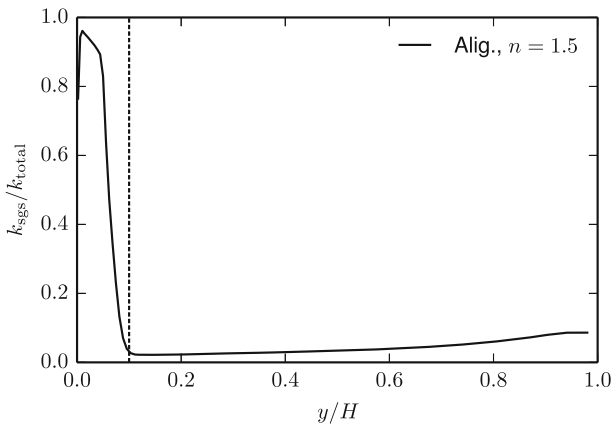


Fig. 4 Profile of the modelled-to-total ratio of the kinetic energy of turbulence for the regular aligned roughness arrangement with $n = 1.5$

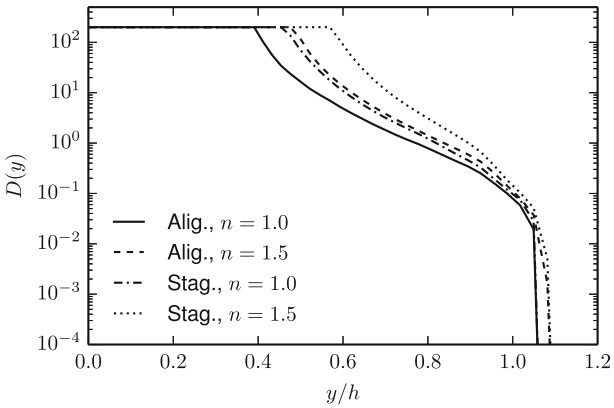


Fig. 5 Drag function profiles for the four considered roughness topographies

the bounding is expected to have a negligible influence on the results. Simulations with smaller bounding values of down to $D = 100$ (not shown here) reinforce this assumption. After exhibiting a high constant (bounded) value in the vicinity of the wall, the drag function decreases and the values approach zero shortly after the roughness elements' crests at $y/h = 1$. Hence, the volumetric forcing term is only active in the immediate vicinity of the roughness layer.

Figure 6 shows the iso-contours of the instantaneous velocity field interpreted in terms of its magnitude in the open channel flow obtained with the present VLES model for the smooth and a rough-wall case (related to aligned roughness arrangement with $n = 1.0$) at an approximately equal friction-based Reynolds number Re_τ. Apart from demonstrating the present model's capability of appropriately resolving the turbulent structures in this flow configuration, Fig. 6 illustrates the influence of the roughness forcing term, resulting in very low velocities in the vicinity of the roughness layer. Consequently, higher velocities in the upper part of the channel are obtained for the rough case, due to the the blockage effects the roughness elements exert on the flow and a subsequent fluid displacement towards to the north boundary plane of the adopted solution domain (note the normalization with the respective bulk velocity). The instantaneous results are averaged over a sufficiently large time interval in order to compare the mean flow field and turbulent intensities to the DNS data base.

The obtained mean velocity profiles in wall units for the case with aligned conic shaped roughness elements as well as the smooth reference case are displayed in Fig. 7. Here, the y coordinate is corrected by the position of the virtual wall y_0, with $y_0 = 0$ in the smooth case. The present model results agree well with the DNS data set for both cases and the

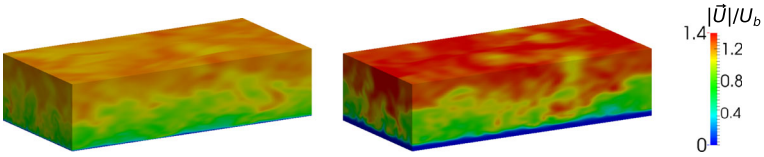


Fig. 6 Instantaneous velocity field coloured by its magnitudes obtained with the present VLES model: comparison of the flow over a smooth (left) and a rough wall (aligned case, $n = 1.0$)

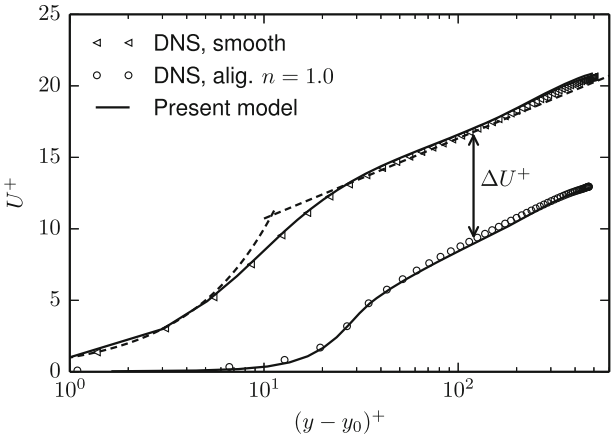
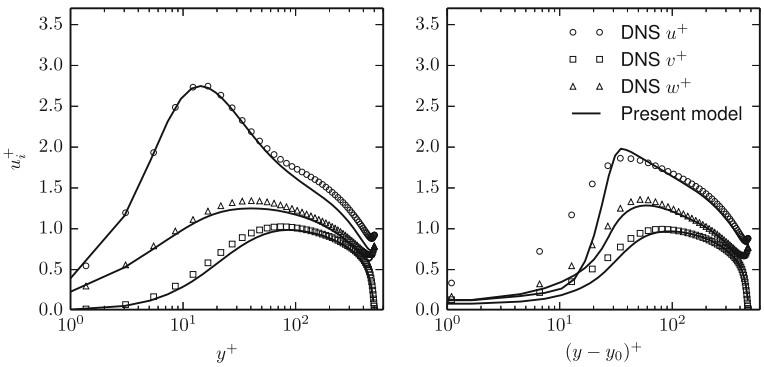


Fig. 7 Semi-log profiles of the mean axial velocity for both the smooth and a rough case (aligned, $n = 1.0$)

typical downward shift ΔU^+ in the logarithmic region due to the roughness presence can be observed.

Normal turbulent intensity components for the smooth reference case are shown in Fig. 8a, while the intensity components for the rough case are displayed in Fig. 8b. The comparison between the DNS data for the smooth and the rough case reveals similar values of the turbulent intensities in the region $(y - y_0)^+ > 100$, which is in agreement with Townsend's [24] outer layer similarity hypothesis. The inner layer however is strongly affected by the roughness, with the streamwise and spanwise turbulent intensity components exhibiting the most significant suppresion, while the wall-normal component is almost not affected or even slightly enhanced near the virtual wall. Hence, the Reynolds-stress anisotropy in the near-wall region is appropriately restructured exhibiting substantial weakening corresponding to the less emphasized separation between normal stress components. Regarding the results with the present VLES model, the turbulent intensities for the smooth case are in a very good agreement with the DNS data. In the rough case, the intensity profiles



(a) Smooth (b) Rough (alig., $n = 1.0$)

Fig. 8 Turbulent intensities of the smooth reference case (**a**) and a rough case (alig. $n = 1.0$) (**b**) in wall units. Legend in (**b**) corresponds to (**a**) as well

in the outer layer are captured correctly. In the near wall region, a slight under-prediction of the wall-normal and spanwise intensity components is obtained. However, the streamwise fluctuations are damped too strongly by the forcing term, resulting in a noticeable under-estimation of the turbulence intensity level. Possible reason for the latter departure is most likely a combination of the following: the proportionality to the square of the velocity might not be completely correct close to the base wall (we recall the value of the effective slope $ES \approx 0.88$ of the presently considered roughness elements), since the total drag is not only due to pressure drag, but also due to viscous drag; deep inside the roughness layer, viscosity effects will play a progressively larger role. Furthermore, the determination of $D(y)$ becomes increasingly difficult closer to the base wall on which the roughness elements are placed, since the velocity, fed into Eq. 14, is very small, resulting eventually in large values of $D(y)$. On the other hand, the focus of the present work is more on correct capturing of the roughness effects on the turbulence level above the roughness crest, since the values inside the roughness layer are an average over a fairly inhomogeneous spatial region.

The corresponding mean velocity profiles normalized by the bulk velocity for all four considered flow cases characterized by the regularly-arranged roughness surface are displayed in Fig. 9. Here as well, the blockage effect exerted by the roughness in the region $y/H < h$ below the roughness crest can be observed. Very good agreement with the DNS data is achieved throughout. Figure 10 shows the velocity profiles nomalized by the friction velocity for the respective rough-wall cases. While U_τ is captured correctly for the first case (alig., $n = 1.0$), some deviations are apparent for the three remaining cases. The relative error in the estimation of U_τ, and correspondingly of Re_τ, is in the range between 1.8 and 8.4% and is primarily due to the insufficiently accurate prediction of the resolved turbulent shear stress component ($\overline{uv}$), which is utilized for the estimation of U_τ using Eq. 17.

Profiles of the normal turbulent intensity components related to the regularly arranged roughness cases, non-dimensionalized by the bulk velocity ($u_i^* = u_i/U_b$), are displayed in Fig. 11. The turbulent structure modification actuated by the rough wall towards the anisotropy weakening is obvious for all four investigated roughness configurations, characterized by a distinct reduction of the peak value of the streamwise intensity component.

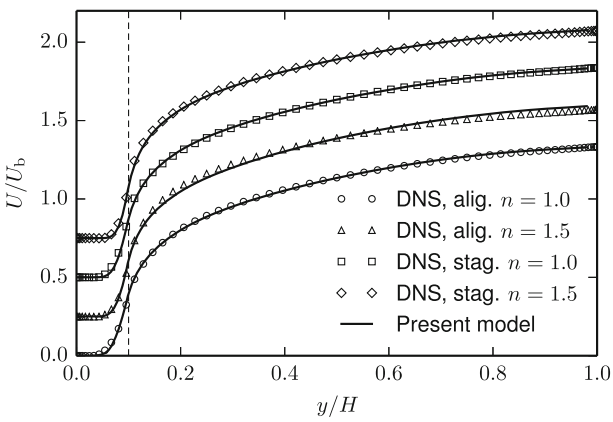


Fig. 9 Velocity profiles normalized by the bulk velocity U_b for the four considered regular roughness topographies. The dashed line represents the crest height of the roughness elements. For better visibility, the profiles are shifted upwards by a value of 0.25, 0.5 and 0.75, respectively

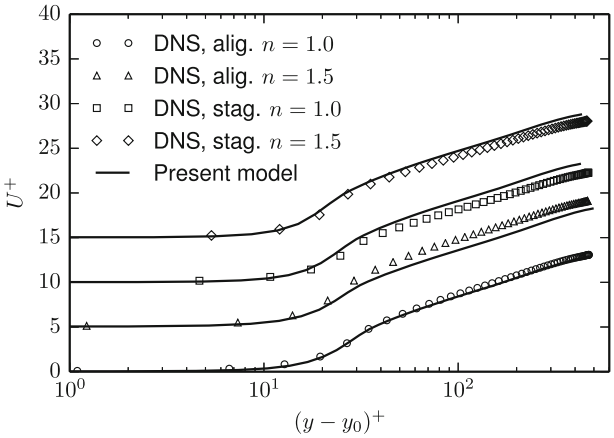


Fig. 10 Semi-log profiles of mean axial velocities for the four considered regular roughness topographies shifted upwards by 5, 10 and 15 units for better visibility

Qualitatively good agreement with the DNS database, with the correctly predicted separation of the turbulent intensities, is achieved outside the roughness layer. Whereas the VLES-related profiles follow closely the DNS results for the aligned cases, in both staggered configurations departure from the DNS results is visible, being especially pronounced in relation to the streamwise component. This is presumably due to a more severe spatial inhomogeneity of the roughness elements arrangement in the streamwise direction, relative to the aligned configurations, which is certainly more challenging for the roughness model because of the inherent assumption of a constant drag distribution in layers parallel to the wall. Clearly, the presently formulated roughness model is not capable of capturing flow features, such as eddy shedding and subsequent reattachment, related to the flow over realistically discretized roughness elements, as it is the case with the complementary DNS simulations. The afore-discussed deviations of the VLES results from the DNS ones can be certainly attributed to this issue. The previously mentioned inaccuracies in the determination of U_τ are closely related to the wall-normal turbulent intensities. Comparison of Figs. 10 and 11 illustrates, that the slight overprediction of v^* in the case of aligned elements with shape parameter $n = 1.5$ is correlated with the overprediction of U_τ, resulting in an underestimation of the semi-log velocity profile normalized by U_τ. The same is valid for the two staggered configurations, where the underprediction of v^* is correlated with the underestimation of U_τ. This seems straightforward, since the production of the turbulent shear stress $\overline{uv}$ is mainly affected by the wall-normal turbulent intensity.

5.2 Randomly arranged roughness

In order to apply the present roughness modeling strategy to a more realistic roughness, a surface with randomly arranged roughness elements was considered. In contrast to the regularly arranged cases, where the arrangement type (aligned/staggered) and element shape ($n = 1.0$, $n = 1.5$) were varied separately, here both, the position and the shape of the roughness elements as well as the peak heights are variably distributed (see Fig. 1-right). On the one hand, this roughness topography exhibits a weaker spatial homogeneity, which can be expected to somewhat relativize the relevance of the present roughness model. But on the other hand, 'geometry-related' flow features such as vortex shedding or flow reattachment,

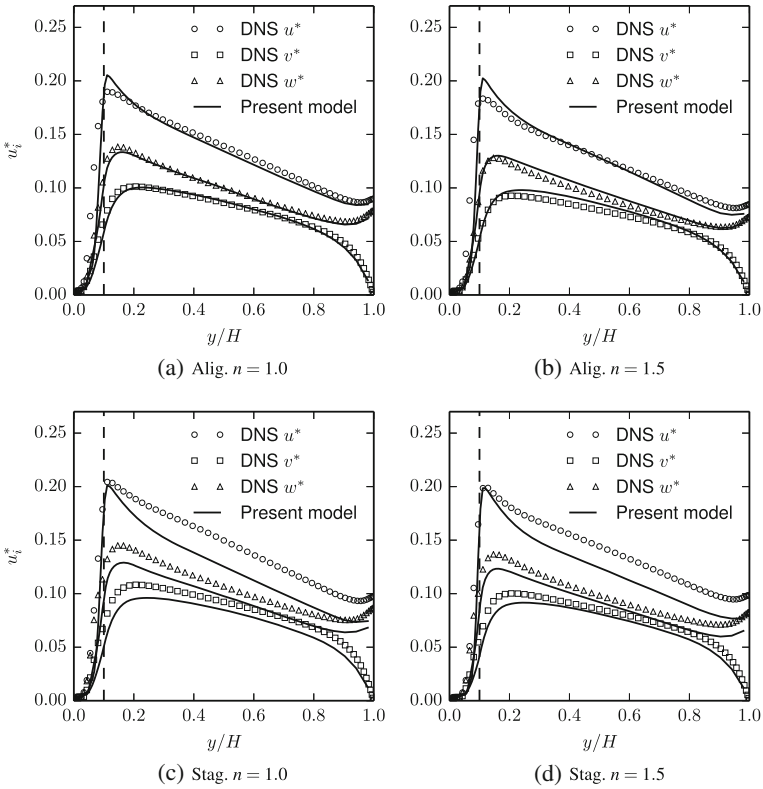


Fig. 11 Turbulent intensity profiles in the channel flow over four regularly-arranged roughness configurations normalized by the bulk velocity - $u_i^* = u_i/U_\mathrm{b}$

which can occur periodically in regularly-arranged cases, are presumably of less importance in a random configuration. Figure 12 displays the mean velocity profile normalized by the bulk velocity. As in the regular roughness cases, the model is capable to correctly capture the mean velocity profile. The relative deviation in the wall-friction velocity is $\Delta U_\tau/U_\tau = 6.7\%$.

Figure 13 shows the turbulent intensities related to the random roughness case normalized by the bulk velocity. The DNS results exhibit some specific differences compared to the regularly-arranged roughness configurations (Fig. 11), characterized by an enhanced peak value of the streamwise component and a somewhat different tendency of the turbulent intensity components by approaching the rough wall - the turbulence level suppression is weaker. In contrast, the VLES results show very similar behaviour as documented in the cases with regular roughness arrangement. Away from the wall ($y/H \geq 0.2$), a slight under-prediction of the intensities is obtained. However, this deviation is less pronounced in comparison with the both regularly-arranged staggered configurations considered previously (Fig. 11c and d). In the near-wall region, the stronger spatial inhomogeneity, mainly affected by the non-uniform peak height in the randomly-arranged case, results in a severe under-estimation of the turbulent intensities compared to the regular roughness cases. This is most likely due to the fact, that in the case of geometrically resolved roughness (as encountered in the reality as well as in the DNS/IBM simulations) the formation of turbulent

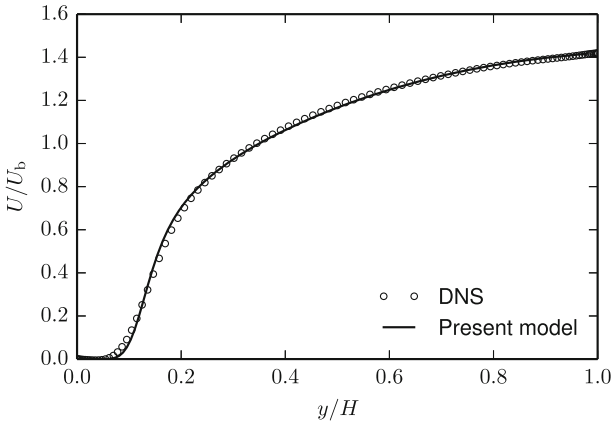


Fig. 12 Velocity profile normalized by the bulk velocity U_b in the channel flow over a randomly arranged roughness elements

structures at a certain wall distance y is not inhibited that much in zones with lower roughness elements compared to zones with larger element heights. In the forcing-based model however, the spatially averaged drag and the associated damping of the resolved turbulent fluctuations is applied independently of the spatial position in streamwise and spanwise directions, but rather uniformly in layers parallel the wall. Consequently, it is more difficult to obtain realistic turbulent structures very close to the virtual wall. A possible route to improvement would be to allow for a full spatial variation of the drag form function in order to capture geometrical variations of the roughness elements larger than the local cell size.

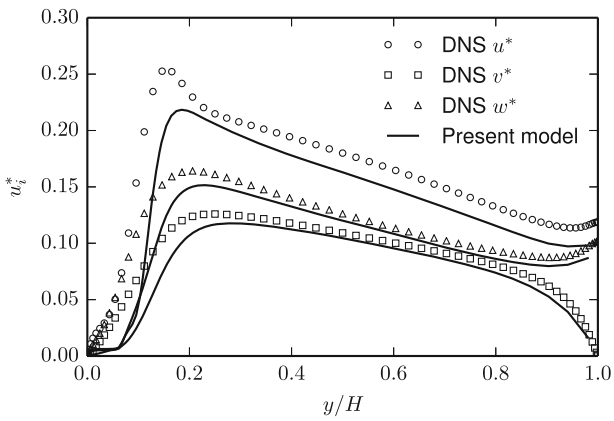


Fig. 13 Turbulent intensity profiles in the channel flow over a randomly-arranged roughness configuration normalized by the bulk velocity - $u_i^* = u_i/U_b$

6 Conclusion

Presently adopted volumetric forcing approach to the modeling of rough surfaces, applied within an eddy-resolving VLES computational framework, relies on introducing an additional drag term in the appropriately 'filtered' Navier-Stokes equations. The latter terms accounts for the form drag and blockage effects, the roughness elements exert on the flow. A non-dimensional drag function $D(y)$ accounting for the shape of roughness elements is introduced. It is determined by 'a priori' evaluating the corresponding RANS equations of motion by utilizing a reference DNS database as input in an open channel flow over variably-shaped rough walls consisting of either a regular or random arrangement of axisymmetric roughness elements. Very good agreement of the mean velocity profiles with the reference DNS data is obtained for all considered roughness arrangements. The global response of the turbulence structure to the wall roughness towards a less anisotropic state is correctly reproduced. However, in some cases the turbulent intensities exhibit departures from the DNS reference, leading to deviations in the friction velocity of up to 8.4%. Further model optimization is necessary to more accurately predict the wall-normal turbulent intensity, which is mainly responsible for correct assessment of the production rate of the turbulent shear stress components, hence affecting directly the estimation of the friction velocity. A possible route is the modification of the drag term, by e.g. giving up the assumption of a wall-parallel uniformity of the drag function, in order to more accurately model the effect of the rough wall on the turbulence. Furthermore, especially in more random roughness configurations with larger variations in the characteristic length scales of the roughness elements or spatially varying roughness 'patches', it might be advantageous to introduce a full three-dimensional spatial dependence of the forcing term.

Acknowledgements The financial support of the German Research Foundation (DFG) in the framework of the Collaborative Research Center/Transregio 150 (TP-B03 and TP-B02) is gratefully acknowledged. The authors furthermore would like to thank for the computing time granted on the Lichtenberg HPC at TU Darmstadt.

Compliance with Ethical Standards

Conflict of interests The authors declare that they have no conflict of interest.

References

1. Bhaganagar, K., Kim, J., Coleman, G.: Effect of roughness in wall-bounded turbulence. Flow, Turbul. Combust. **72**, 463–492 (2004)
2. Nikuradse, J.: Strömungsgesetze in rauen Rohren. VDI-Forschungshefte Band 361 (1933)
3. Taylor, R.P., Coleman, H.W., Hodge, B.K.: Prediction of turbulent rough-wall skin friction using a discrete element approach. ASME J. Fluids Eng. **107**, 251–257 (1985)
4. Miyake, Y., Tsujimoto, K., Agata, Y.: A DNS of a turbulent flow in a rough-wall channel using roughness elements model. JSME Int. J. Series B **43**(2), 233–242 (2000)
5. Busse, A., Sandham, N.D.: Parametric forcing approach to rough-wall turbulent channel flow. J. Fluid Mech. **712**, 169–202 (2012)
6. Forooghi, P., Stroh, A., Magagnato, F., Jakirlic, S., Frohnapfel, B.: Towards a universal roughness correlation. ASME J. Fluids Eng. **139**(12), 121201–12201-12 (2017)
7. Tarada, F.: Prediction of rough-wall boundary layers using a low Reynolds number $k-\varepsilon$ model. Int. J. Heat Fluid Flow **11**(4), 331–345 (1990)
8. Stripf, M., Schulz, A., Bauer, H.-J., Wittig, S.: Extended models for transitional rough wall boundary layers with heat transferpart I: model formulations. J. Turbomach. **131**(3), 031016 (2009)

9. Bons, J.P., Taylor, R.P., McClain, S.T., Rivir, R.B.: The many faces of turbine surface roughness. J. Turbomach. **123**(4), 739–748 (2001)
10. Speziale, C.G.: Turbulence modeling for time-dependent RANS and VLES: A review. AIAA J. **36**(2), 173–184 (1998)
11. Schiestel, R., Dejoan, A.: Towards a new partially integrated transport model (PITM) for coarse grid and unsteady turbulent flow simulations. Theoret. Comput. Fluid Dyn. **18**(6), 443–468 (2005)
12. Basara, B., Krajnovic, S., Girimaji, S., Pavlovic, Z.: Near-wall formulation of the partially averaged navier-stokes (PANS) turbulence model. AIAA J. **49**(12), 2627–2636 (2011)
13. Hanjalić, K., Popovac, M., Hadžiabdić, M.: A robust near-wall elliptic-relaxation eddy-viscosity turbulence model for CFD. Int. J. Heat Fluid Flow **25**, 1047–1051 (2004)
14. Chang, C.-Y., Jakirlić, S., Dietrich, K., Basara, B., Tropea, C.: Swirling flow in a tube with variably-shaped outlet orifices: an LES and VLES study. Int. J. Heat Fluid Flow **49**, 28–42 (2014)
15. Chang, C.-Y., Krumbein, B., Jakirlic, S., Tropea, C., Basara, B., Sadiki, A., Janicka, J., Böhm, B., Dreizler, A., Peterson, B.: Flow dynamics in IC-engine configurations simulated by scale-resolving models. International Multidimensional Engine Modeling (IMEM) User's Group Meeting at the SAE Congress, Detroit, MI (2016)
16. Jakirlic, S., Kutej, L., Hanssmann, D., Basara, B., Tropea, C.: Eddy-resolving Simulations of the Notchback DrivAer Model: Influence of Underbody Geometry and Wheels Rotation on Aerodynamic Behaviour. SAE Technical Paper Series, Paper No. 2016-01-1062 (2016)
17. Aupoix, B.: Improved heat transfer predictions on rough surfaces. Int. J. Heat Fluid Flow **56**, 160–171 (2015)
18. Belcher, S.E., Jerram, N., Hunt, J.C.R.: Adjustment of a turbulent boundary layer to a canopy of roughness elements. J. Fluid Mech. **488**, 369–398 (2003)
19. Stripf, M., Schulz, A., Bauer, H.-J.: Modeling of rough-wall boundary layer transition and heat transfer on turbine airfoils. J. Turbomach. **130**(2), 021003 (2008)
20. Ashravian, A., Andersson, H.I., Manhart, M.: DNS of turbulent flow in a rod-roughened channel. Int. J. Heat Fluid Flow **25**, 373–383 (2004)
21. Chevalier, M., Schlatter, P., Lundbladh, A., Henningson, D.S.: SIMSON – A pseudo-spectral solver for incompressible boundary layer flow. KTH Mechanics, Stockholm (2007)
22. Goldstein, D., Handler, R., Sirovich, L.: Modeling a no-slip flow boundary with an external force field. J. Comput. Phys. **105**(2), 354–366 (1993)
23. Chan-Braun, C., Garcia-Villalba, M., Uhlmann, M.: Force and torque acting on particles in a transitionally rough open-channel flow. J. Fluid Mech. **684**, 441–474 (2011)
24. Townsend, A.A. The structure of of turbulent shear flow, 2nd edn. Cambridge University Press, Cambridge (1976)